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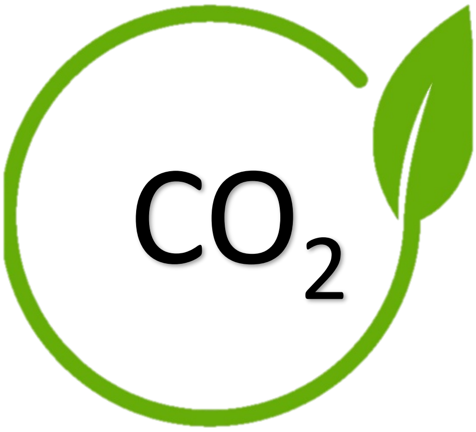
WHAT ARE CARBON CREDITS AND HOW DO THEY WORK?



What are Carbon Credits and how do they work?

A **carbon credit represents one tonne of carbon dioxide reduced, avoided, or removed from the atmosphere** through any project.

For example:

•If the government launches a **tree plantation project** which cuts **10 tonnes of carbon**, it can earn **10 carbon credits**.

•The government can then **sell these credits in the carbon market** to a company, which wants to show reduction in carbon emissions.

If a company is **struggling to meet its emission reduction target for a year**, it can buy these carbon credits and claim to have met the target.

Key insight:

Decarbonisation is an **expensive process**. Governments or companies have **limited funds** to spend.

Example:

•It may cost the government **more money to buy electric buses** than diesel buses for public transport.

•But electric buses are **better for the environment**.

Carbon trading makes such activities financially viable, says **Vaibhav Chaturvedi**, Senior Fellow, **Council on Energy, Environment and Water (CEEW)**.

How do carbon markets function and what is the price of a credit?

•Companies or governments **voluntarily buy carbon credits** as part of:

- **Corporate Social Responsibility (CSR)**, or
- **Compliance with government regulations**

•The existence of **global, national, and sub-national goals for carbon emission reduction** means there will be **buyers** (companies struggling to meet emission standards) who want to buy carbon credits.

Global scenario

•There are around **80 carbon trading regimes worldwide**, according to the **World Bank's State and Trends of Carbon Pricing 2025 report**.

•Some markets are run by **international non-profit organisations**, others by **governments**.

•**India is currently in the process of setting up its own carbon market**.

Verification

•**Private third-party firms**, accredited by these markets, **audit and verify emission reductions** of companies or projects meticulously.

Price of carbon credits

•The **value of a carbon credit varies from $1 to over $100**, depending on **market dynamics**.

- Price depends on:
    - **Demand and supply of credits**
    - **Type and quality of credit**

According to the World Bank report:

- **28% of global emissions will be covered by carbon pricing in 2025**, up from **just 5% in 2005**
- **Revenues worth over $100 billion were raised in 2025 alone**